

STAT 516: Homework 8

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. **(10 points)** Let X be a continuous random variable with density function, for $b > 0$,

$$f(x) = \begin{cases} be^{-bx}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

The MGF of X is denoted by $M_X(t)$. What is $M_X(-6b)$?

2. **(13 points)** Let X have the PMF

$$P_X(x) = \begin{cases} \left(\frac{8}{9}\right)^x \left(\frac{1}{9}\right), & x = 0, 1, 2, \dots, \\ 0, & \text{otherwise.} \end{cases}$$

What is the MGF of X ?

3. **(12 points)** The moment generating function for X is

$$M(t) = c + 0.4e^t + 0.3e^{2t} + 0.2e^{at},$$

where c is a constant. The mean of the variable is 2. Determine a .

4. **(13 points)** Let the random variable X have moment generating function

$$M(t) = e^{3t+t^2}.$$

What is $E[X^2]$?

5. **(13 points)** Let X be a random variable with moment generating function

$$M(t) = \left(\frac{2 + e^t}{3}\right)^9.$$

Calculate the variance of X .

6. **(13 points)** An actuary determines that the claim size for a certain class of accidents is a random variable X with moment generating function

$$M(t) = \frac{1}{(1 - 2500t)^4}.$$

Calculate the standard deviation of the claim size for this class of accidents.

7. **(13 points)** Two claimants place calls simultaneously to an insurer's claims call center. The times X and Y , in minutes, that elapse before the respective claimants get to speak with call center representatives are independently and identically distributed. The moment generating function of each random variable is

$$M(t) = \frac{1}{(1 - 1.5t)^2}, \quad t < \frac{2}{3}.$$

Calculate the standard deviation of $X + Y$.

8. **(13 points)** The moment generating function of X is

$$M(t) = \frac{1}{(1 - 0.1t)^3}.$$

Let $Y = 3X + 5$. Let $M_Y(t)$ be the moment generating function of Y . Determine $M_Y(0.5)$.

1. (10 points) Let X be a continuous random variable with density function, for $b > 0$,

$$f(x) = \begin{cases} be^{-bx}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Exponential Function with rate b . $X \sim \text{Exp}(b)$

The MGF of X is denoted by $M_X(t)$. What is $M_X(-6b)$?

$$M_X(t) = \frac{b}{b-t}, \quad t < b$$

$$t = -6b: M_X(t) = \frac{b}{b-(-6b)} = \frac{b}{7b}$$

$$\text{Since } b > 0, \quad M_X(-6b) = 1/7$$

2. (13 points) Let X have the PMF

$$P_X(x) = \begin{cases} \left(\frac{8}{9}\right)^x \left(\frac{1}{9}\right), & x = 0, 1, 2, \dots, \\ 0, & \text{otherwise.} \end{cases}$$

Geometric - type $p = 1/9$

What is the MGF of X ?

$$M_X(t) = E(e^{tx})$$

$$M_X(t) = \sum_{x=0}^{\infty} e^{tx} P_X(x) \longrightarrow \sum_{x=0}^{\infty} e^{tx} \left(\frac{8}{9}\right)^x \left(\frac{1}{9}\right) = \frac{1}{9} \sum_{x=0}^{\infty} \left(\frac{8}{9} e^t\right)^x$$

$$\text{Geometric Series: } \sum_{x=0}^{\infty} r^x = \frac{1}{1-r}, \quad |r| < 1$$

$$\text{Therefore: } M_X(t) = \frac{1}{9} \cdot \frac{1}{1 - \frac{8}{9}e^t} = \frac{1}{9 - 8e^t}$$

$$M_X(t) = \frac{1}{9 - 8e^t}, \quad t < \ln\left(\frac{9}{8}\right)$$

$$\begin{aligned} & \downarrow \\ & \left| \frac{8}{9} e^t \right| < 1 \\ & \text{Since } e^t > 0: \frac{8}{9} e^t < 1 \rightarrow e^t < \frac{9}{8} \\ & t < \ln\left(\frac{9}{8}\right) \end{aligned}$$

3. (12 points) The moment generating function for X is

$$M(t) = c + 0.4e^t + 0.3e^{2t} + 0.2e^{at},$$

where c is a constant. The mean of the variable is 2. Determine a .

$$E[X] = 2$$

$$E[X] = M'_X(0)$$

$$M'_X(t) = 0.4e^t + 0.6e^{2t} + 0.2ae^{at}$$

$$M'_X(0) = 0.4 + 0.6 + 0.2a = 2$$

$$1 + 0.2a = 2$$

$$0.2a = 1 \quad a = 5$$

4. (13 points) Let the random variable X have moment generating function

$$M(t) = e^{3t+t^2}.$$

What is $E[X^2]$?

$$E[X^2] = M''_X(0)$$

$$M'_X(t) = (3+2t)e^{3t+t^2}$$

$$M''_X(t) = 2e^{3t+t^2} + (3+2t)^2 e^{3t+t^2} = [2 + (3+2t)^2] e^{3t+t^2}$$

$$M''_X(0) = [2 + 3^2] e^0 = 11, \text{ Hence } E[X^2] = 11$$

5. (13 points) Let X be a random variable with moment generating function

$$M(t) = \left(\frac{2+e^t}{3} \right)^9 = \left(\frac{2}{3} + \frac{1}{3}e^t \right)^9$$

Calculate the variance of X .

Binomial

$$\text{Binomial MGF} = M_X(t) = [(1-p) + pe^t]^n$$

$$n=9, p=\frac{1}{3} \rightarrow X \sim \text{Bin}(9, \frac{1}{3})$$

$$\text{Binomial RV, } \text{Var}(X) = np(1-p) \rightarrow 9\left(\frac{1}{3}\right)\left(1-\frac{1}{3}\right) = 9\left(\frac{1}{3}\right)\left(\frac{2}{3}\right) = 2$$

$$\text{Var}(X) = 2$$

6. (13 points) An actuary determines that the claim size for a certain class of accidents is a random variable X with moment generating function

$$M(t) = \frac{1}{(1-2500t)^4}.$$

Gamma RV

Shape α , Rate β

SD(X)

Calculate the standard deviation of the claim size for this class of accidents.

$$\text{Gamma MGF} = M_X(t) = \left(\frac{\beta}{\beta-t} \right)^\alpha = \left(1 - \frac{t}{\beta} \right)^{-\alpha} \rightarrow (1-2500t)^{-4}$$

$$\begin{aligned} \alpha &= 4 \\ \frac{1}{\beta} &= 2500 \\ \beta &= 1/2500 \end{aligned}$$

$$\text{Gamma RV, } \text{Var}(X) = \alpha/\beta^2 \rightarrow 4/(1/2500)^2 = 4(2500)^2 = 25,000,000$$

$$\text{SD}(X) = \sqrt{25,000,000}$$

$$\text{SD}(X) = 5000$$

7. (13 points) Two claimants place calls simultaneously to an insurer's claims call center. The times X and Y , in minutes, that elapse before the respective claimants get to speak with call center representatives are independently and identically distributed. The moment generating function of each random variable is

$$M(t) = \frac{1}{(1 - 1.5t)^2}, \quad t < \frac{2}{3}.$$

Calculate the standard deviation of $X + Y$.

Gamma RV: shape α Scale θ

$$\alpha = 2, \theta = 1.5, \beta = \frac{1}{1.5} = \frac{2}{3}$$

$$\text{Var}(X) = \alpha \theta^2 = 2(1.5)^2 = 4.5$$

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) = 4.5 + 4.5 = 9$$

$$\text{SD}(X+Y) = \sqrt{9}$$

$$\text{SD}(X+Y) = 3$$

8. (13 points) The moment generating function of X is

$$M(t) = \frac{1}{(1 - 0.1t)^3}.$$

Let $Y = 3X + 5$. Let $M_Y(t)$ be the moment generating function of Y . Determine $M_Y(0.5)$.

Linear Transformation $Y = aX + b$:

$$M_Y(t) = E(e^{tY}) = E(e^{t(ax+b)}) = e^{bt} E(e^{atX}) = e^{bt} M_X(at)$$

$$M_Y(t) = e^{5t} M_X(3t)$$

$$M_X(3t) = \frac{1}{(1 - 0.1(3t))^3} = \frac{1}{(1 - 0.3t)^3}$$

$$M_Y(t) = e^{5t} / (1 - 0.3t)^3$$

$$M_Y(0.5) = e^{5(0.5)} / (1 - 0.3(0.5))^3 = \frac{e^{2.5}}{(1 - 0.15)^3} = \frac{e^{2.5}}{0.85^3}$$

$$M_Y(0.5) \approx 19.84$$